THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 16 **(b)** 14 **(c)** 15 **(d)** 17

2. 2^7 added 2 times equals: 1

(a) 2^9 **(b)** 2^{14} **(c)** 2^8 **(d)** 4^7

3. If $(34)_b = 19$ in decimal (a two-digit number in base b), then b equals: 1

(a) 7 **(b)** 6 **(c)** 4 **(d)** 5

4. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 102° **(b)** 90° **(c)** 144° **(d)** 198°

5. The smallest number which leaves remainder 5 when divided by 12 and 18 is: 1

(a) 41 **(b)** 221 **(c)** 36 **(d)** 77

6. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 109 **(b)** 67 **(c)** 53 **(d)** 81

7. If x : y = 2 : 3 and y : z = 4 : 7, then x : y : z is: 1

(a) 2 : 3 : 7 **(b)** 2 : 4 : 7 **(c)** 8 : 12 : 21 **(d)** 2 : 7 : 7

8. Two articles are sold at ₹ 1800 each, one at a gain of 10% and the other at a loss of 10%. The overall result is a loss of (in %): 1

(a) 0% **(b)** 10% **(c)** 5% **(d)** 1%

9. The legs of a right triangle are in the ratio $5 : 12$ and its hypotenuse is 52 cm. Its perimeter (in cm) is: 1

(a) 104 **(b)** 124 **(c)** 120 **(d)** 68

10. The present ages of Ria and Sia are in the ratio 3:5. After 8 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 32 **(b)** 40 **(c)** 22 **(d)** 24

11. All the 12 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 18 **(b)** 30 **(c)** 20 **(d)** 44

12. The mean of 5 numbers is 30. When one number is removed the mean of the remaining 4 becomes 27. The removed number is: 1

(a) 3 **(b)** 30 **(c)** 15 **(d)** 42

13. A two-digit number exceeds the number formed by reversing its digits by 18. The difference of its digits is: 1

(a) 18 **(b)** 2 **(c)** 1 **(d)** 3

14. The perimeter of a rectangle is 128 cm and the length is 3 times the breadth. Its area is: 1

(a) 784 cm² **(b)** 1024 cm² **(c)** 736 cm² **(d)** 768 cm²

15. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 61 **(b)** 64 **(c)** 63 **(d)** 62

16. If $2^{2x+1} = 2^9$, the value of x is: 1

(a) 5 **(b)** 4 **(c)** 3 **(d)** 8

17. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 31 kg is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

18. One angle of a parallelogram is 2 times its adjacent angle. The smaller angle is: 1

(a) 80° **(b)** 90° **(c)** 60° **(d)** 120°

19. The number of multiples of 7 between 100 and 500 (both included) is: 1

(a) 59 **(b)** 58 **(c)** 57 **(d)** 56

20. If $x - y = 3$ and $xy = 20$, then $x^2 + y^2$ equals: 1

(a) 49 **(b)** 9 **(c)** -31 **(d)** 29

21. The present ages of two brothers are in the ratio 3 : 2. After 4 years the ratio becomes 7 : 5. The present age of the elder brother is: 1

(a) 16 years **(b)** 28 years **(c)** 24 years **(d)** 40 years

22. The population of a town is 50000 and it grows at 10% per annum. After 2 years it will be: 1

(a) 54450 **(b)** 60500 **(c)** 60000 **(d)** 66550

23. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 45 cm. The area of the rectangle (in cm²) is: 1

(a) 126 **(b)** 405 **(c)** 972 **(d)** 1053

24. ₹ 4500 is divided among three friends in the ratio 2:3:5. The difference between the largest and the smallest share (in ₹) is: 1

(a) 2250 **(b)** 1350 **(c)** 900 **(d)** 1351

25. A polyhedron has 18 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

(a) 6 **(b)** 8 **(c)** 20 **(d)** 12

26. The mean of 10 values was found to be 36. Later it was found that 80 was misread as 30. The correct mean is: 1

(a) 31 **(b)** 36 **(c)** 41 **(d)** 86

27. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 11 **(b)** 7 **(c)** 6 **(d)** 9

28. An isosceles trapezium has parallel sides 10 cm and 22 cm and non-parallel sides 10 cm. Its area is: 1

(a) 160 cm² **(b)** 128 cm² **(c)** 256 cm² **(d)** 138 cm²

29. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 79 **(b)** 77 **(c)** 80 **(d)** 78

30. The number of digits in $2^4 \times 5^7$ is: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

31. Which of the following numbers is the greatest? 1

(a) 14_8 **(b)** 32_5 **(c)** 1101_2 **(d)** 16_{10}

32. In a rhombus ABCD, $\angle ACD = 25^\circ$. Then $\angle ABC$ is: 1

(a) 130° **(b)** 50° **(c)** 155° **(d)** 65°

33. The number of 3-digit numbers divisible by 8 is: 1

(a) 112 **(b)** 113 **(c)** 111 **(d)** 124

34. The value of $512^2 - 488^2$ is: 1

(a) 24100 **(b)** 24000 **(c)** 500288 **(d)** 576

35. ₹ 1300 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. The largest part is: 1

(a) ₹ 400 **(b)** ₹ 300 **(c)** ₹ 600 **(d)** ₹ 260

36. A machine worth ₹ 50000 depreciates at 10% per annum. Its value after 2 years (in ₹) is: 1

(a) 40500 **(b)** 44550 **(c)** 40000 **(d)** 36450

37. A Pythagorean triplet whose smallest member is 15 is: 1

(a) 15, 113, 114 **(b)** 15, 16, 17 **(c)** 15, 112, 113 **(d)** 15, 111, 112

38. In a pie chart a 45° sector represents 13 students. The total number of students is: 1

(a) 104 **(b)** 117 **(c)** 468 **(d)** 52

39. If every corner of a cube is sliced off (without removing the whole edges), the number of vertices of the resulting solid is: 1

(a) 12 **(b)** 16 **(c)** 24 **(d)** 8

40. A class has 29 students with mean age 13 years. When the teacher's age is included, the mean rises to 14. The teacher's age (years) is: 1

(a) 38 **(b)** 43 **(c)** 14 **(d)** 42

41. **Assertion (A):** The number 1296 is a perfect square. 1

Reason (R): A number is a perfect square if every prime in its prime factorisation occurs an even number of times.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** $2^{-3} = \frac{1}{8}$. 1

Reason (R): For a non-zero number a, $a^{-n} = \frac{1}{a^n}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** The binary number 101_2 equals 5 in decimal. 1

Reason (R): In base 2, the value of a number is the sum of each digit times a power of 2, with the rightmost place being 2^0.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** The adjacent angles of a parallelogram are supplementary. 1

Reason (R): The opposite sides of a parallelogram are parallel, so adjacent angles are co-interior angles on a transversal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** The number 4,236 is divisible by 3. 1
Reason (R): A number is divisible by 3 if the sum of its digits is divisible by 3.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** $(x+5)^2 = x^2 + 10x + 25$. 1
Reason (R): $(a+b)^2 = a^2 + 2ab + b^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The ratios $3:4$ and $9:12$ are in proportion. 1
Reason (R): Two ratios $a:b$ and $c:d$ are in proportion if they are equal, that is, if $ad = bc$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** 25% of 360 is 90. 1
Reason (R): $x\%$ of a quantity N equals $\frac{x}{100} \times N$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A triangle with sides 6 cm, 8 cm and 10 cm is right-angled. 1
Reason (R): If the square of the longest side equals the sum of the squares of the other two sides, the triangle is right-angled.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** If 5 kg of rice costs ₹300, then 8 kg of rice costs ₹480. 1
Reason (R): The cost of rice is directly proportional to its quantity, so $\frac{\text{cost}}{\text{quantity}}$ is constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**